

SQL Lab Exercises #3

SELECT Queries (Based on Rajagiri Academic System)

Part A — Basic SELECT Queries

1. Display all records from the DEPARTMENT table.

Select * from department1;

```
SQL> select * from department1;
```

DEPT_ID	DEPT_NAME	OFFICE_LOCATION	CONTACT_NUMBER	HOD_FACULTY_ID
D01	Computer Science	Technology Block	9876543210	
D02	Electronics	Science Block		
D03	Mathematics	North Block	9876501111	
D04	Physics	East Block	9876502222	
D05	Commerce	South Block	9876503333	

2. Display all records from the STUDENT table.

SQL> select * from student1;

```
SQL> SET LINESIZE 500;
SQL> SET PAGESIZE 50;
SQL> select * from student1;
```

STUDENT_ID	STUDENT_NAME	ADMISSION	CGPA	CITY	STATE	CONTACT_NUMBER	DOB	G	EMAIL_ID	DEPT_ID
S101	Akhil Raj	26-JUL-26	8.9			9876543210	15-AUG-05	m	s101@gmail.com	D01
S102	Arun Kumar	26-JUL-26				9876545678	20-JAN-05	m	arun@gmail.com	D01
S103	Anjali Nair	26-JUL-26	9.6			9876512345	12-MAR-05	f	anjali@gmail.com	D01

3. Display all records from the FACULTY table.

SQL> select * from faculty1;

```
SQL> select * from faculty1;
```

FACULTY_ID	FACULTY_NAME	DESIGNATION	JOINING_D	SALARY	EMAIL_ID	DEPT_ID
EXPERIENCE_YEARS						
F101	Meera Joseph	Assistant Professor	01-JUN-22	71280	meera@rajagiri.edu	D01
F102	Rahul Menon	Assistant Professor	15-JUL-21	59400	rahul@rajagiri.edu	D01
F103	Anitha Nair	Professor	10-JUN-20	77000	anitha@rajagiri.edu	D02

4. Display all records from the COURSE table.

select * from course1;

```
SQL> _
```

COURSE_ID	COURSE_TITLE	CREDITS	SEMESTER	DEPT_ID	FACULTY_ID
C101	DBMS	5	3	D01	F102
C102	Operating Systems	4	3	D01	F102

5. Display all records from the ENROLLMENT table.

SQL> select * from enrollment1;

ENROLLMENT_ID	STUDENT_ID	COURSE_ID	ENROLLMENT_DATE	ACADEMIC_YEAR
E101	S101	C101	26-JUL-26	2026-2027

6. Display the student ID, student name, gender, and CGPA of all students.

Select * from faculty

select STUDENT_ID,STUDENT_NAME,gender,cgpa from student1

STUDENT_ID	STUDENT_NAME	G	CGPA
S101	Akhil Raj	m	8.9
S102	Arun Kumar	m	
S103	Anjali Nair	f	9.6

7. Display the faculty ID, faculty name, designation, and salary of all faculty members.

SQL> select FACULTY_ID, FACULTY_NAME, DESIGNATION ,SALARY FROM FACULTY1;

FACULTY_ID	FACULTY_NAME	DESIGNATION	SALARY
F101	Meera Joseph	Assistant Professor	71280
F102	Rahul Menon	Assistant Professor	59400
F103	Anitha Nair	Professor	77000

8. Display the course ID, course name, credits, and semester of all courses.

SQL> select COURSE_ID, COURSE_TITLE , credits, semester from course1;

COURSE_ID	COURSE_TITLE	CREDITS	SEMESTER
C101	DBMS	5	3
C102	Operating Systems	4	3

9. Display the event name, event date, venue, and registration fee of all events.

SQL> select EVENT_ID, EVENT_NAME,EVENT_DATE, VENUE, REGISTRATION_FEE from event1;

EVENT_ID	EVENT_NAME	EVENT_DATE	VENUE	REGISTRATION_FEE
EV101	Tech Fest	15-AUG-26	Main Auditorium	500
EV102	Sports Day	20-SEP-26	College Ground	15
EV103	Cultural Fest	10-OCT-26	Open Stage	315

10. Display the result ID, internal marks, external marks, total marks, grade, and result status from the RESULT table.

```
SQL> SELECT * FROM RESULT1;

no rows selected
```

Part B — DISTINCT, Aliases, and Calculated Columns

11. Display the different faculty designations without duplicates.

```
SQL> select distinct designation from faculty1;
```

```
DESIGNATION
-----
Assistant Professor
Professor
```

12. Display the different qualifications available in the FACULTY table.

```
SQL> SELECT DISTINCT QUALIFICATION
  2  FROM FACULTY1;
SELECT DISTINCT QUALIFICATION
                *

ERROR at line 1:
ORA-00904: "QUALIFICATION": invalid identifier
```

13. Display the different semesters in which courses are offered.

```
SQL> select distinct semester from course1;
```

```
SEMESTER
-----
3
```

14. Display the student name and CGPA using the aliases Student_Name and Grade_Point_Average.

SQL> select STUDENT_NAME as Student_Name,cgpa as Grade_Point_Average from student1;

STUDENT_NAME	GRADE_POINT_AVERAGE
Akhil Raj	8.9
Arun Kumar	
Anjali Nair	9.6

15. Display the faculty name, monthly salary, and annual salary. Calculate annual salary as monthly salary multiplied by 12.

SQL> SELECT FACULTY_NAME, SALARY AS MONTHLY_SALARY, SALARY * 12 AS ANNUAL_SALARY FROM FACULTY1;

FACULTY_NAME	MONTHLY_SALARY	ANNUAL_SALARY
Meera Joseph	71280	855360
Rahul Menon	59400	712800
Anitha Nair	77000	924000

16. Display the faculty name, current salary, and salary after a 10% increase without modifying the table.

SQL> select FACULTY_NAME, salary ,salary * 1.10 as salary_after_increase from faculty1;

FACULTY_NAME	SALARY	SALARY_AFTER_INCREASE
Meera Joseph	71280	78408
Rahul Menon	59400	65340
Anitha Nair	77000	84700

17. Display the result ID, internal marks, external marks, and a calculated total obtained by adding the two marks.

SELECT RESULT_ID,
INTERNAL_MARKS,
EXTERNAL_MARKS,
INTERNAL_MARKS + EXTERNAL_MARKS AS TOTAL
FROM RESULT;

18. Display the event name, registration fee, and registration fee after adding 18% tax.

SQL> select EVENT_NAME, REGISTRATION_FEE, REGISTRATION_FEE *1.18 from event1;

EVENT_NAME	REGISTRATION_FEE	REGISTRATION_FEE*1.18
Tech Fest	500	590
Sports Day	15	17.7
Cultural Fest	315	371.7

Part C — WHERE Clause and Relational Operators

19. Display the details of all female students.

```
SQL> select * from student1 where gender ='f';
```

STUDENT_ID	STUDENT_NAME ADMISSION	CGPA CITY	STATE	CONTACT_NUMBER	DOB	G EMAIL_ID	DEPT_ID
S103	Anjali Nair 26-JUL-26	9.6		9876512345	12-MAR-05	f anjali@gmail.com	D01

20. Display the details of students whose CGPA is greater than 8.

```
SQL> select * from student1 where cgpa >8;
```

STUDENT_ID	STUDENT_NAME ADMISSION	CGPA CITY	STATE	CONTACT_NUMBER	DOB	G EMAIL_ID	DEPT_ID
S101	Akhil Raj 26-JUL-26	8.9		9876543210	15-AUG-05	m s101@gmail.com	D01
S103	Anjali Nair 26-JUL-26	9.6		9876512345	12-MAR-05	f anjali@gmail.com	D01

21. Display students whose CGPA is less than or equal to 6.

```
SQL> select * from student1 where cgpa <=6;

no rows selected
```

22. Display students belonging to department D01.

```
SQL> select * from student1 where dept_id ='D01';
```

STUDENT_ID	STUDENT_NAME	ADMISSION	CGPA	CITY	STATE	CONTACT_NUMBER	DOB	G E/MAIL_ID	DEPT_ID
S101	Akhil Raj	26-JUL-26	8.9			9876543210	15-AUG-05 m	s101@gmail.com	D01
S102	Arun Kumar	26-JUL-26				9876545678	20-JAN-05 m	arun@gmail.com	D01
S103	Anjali Nair	26-JUL-26	9.6			9876512345	12-MAR-05 f	anjali@gmail.com	D01

23. Display faculty members whose salary is greater than ₹50,000.

SQL> SELECT * FROM FACULTY1 WHERE SALARY >50000;

FACULTY_ID	FACULTY_NAME	DESIGNATION	JOINING_D	SALARY	E/MAIL_ID	DEPT_ID	EXPERIENCE_YEARS
F101	Meera Joseph	Assistant Professor	01-JUN-22	71280	meera@rajagiri.edu	D01	
F102	Rahul Menon	Assistant Professor	15-JUL-21	59400	rahul@rajagiri.edu	D01	
F103	Anitha Nair	Professor	10-JUN-20	77000	anitha@rajagiri.edu	D02	

24. Display faculty members whose designation is Assistant Professor.

SQL> select faculty_name from faculty1 where designation ='Assistant Professor';

FACULTY_NAME
Meera Joseph
Rahul Menon

25. Display faculty members whose qualification is PhD.

**SELECT FACULTY_NAME
FROM FACULTY1
WHERE QUALIFICATION = 'phd';**

26. Display courses offered in Semester 3.

COURSE_ID	COURSE_TITLE	CREDITS	SEMESTER	DEPT_ID	FACULTY_ID
C101	DBMS	5	3	D01	F102
C102	Operating Systems	4	3	D01	F102

27. Display courses having more than four credits.

```
SQL> select * from course1 where credits>4;
```

COURSE_ID	COURSE_TITLE	CREDITS	SEMESTER	DEPT_ID	FACULTY_ID
C101	DBMS	5	3	D01	F102

28. Display all results with result status PASS.

Select * from result1 where result_status ='pass';

```
SQL> select * from result1 where result_status ='PASS';
```

RESULT_ID	ENROLLMENT_ID	INTERNAL_MARKS	EXTERNAL_MARKS	TOTAL_MARKS	G	RESULT_STA
R101	E101	28	45	73	B	PASS
R102	E102	35	42	77	B	PASS
R103	E103	40	48	88	A	PASS

29. Display results in which total marks are greater than or equal to 80.

Select * from result1 where total_marks >=80;

RESULT_ID	ENROLLMENT_ID	INTERNAL_MARKS	EXTERNAL_MARKS	TOTAL_MARKS	G	RESULT_STA
R103	E103	40	48	88	A	PASS

30. Display results in which internal marks are greater than external marks.

Select * from result1 where internal_marks>external_marks;

```
SQL> Select * from result1 where internal_marks >external_marks;
```

no rows selected

31. Display events for which the registration fee is greater than ₹500.

```
SQL> select * from event1 where REGISTRATION_FEE>500;

no rows selected
```

32. Display events that can accommodate more than 100 participants.

```
SQL> select * from event1 where MAX_PARTICIPANTS>100;
```

EVENT_ID	EVENT_NAME	EVENT_DATE	VENUE	FACULTY_ID	MAX_PARTICIPANTS	REGISTRATION_FEE
EV102	Sports Day	20-SEP-26	College Ground	F102	200	15
EV103	Cultural Fest	10-OCT-26	Open Stage	F103	150	315

33. Display event registrations with participation status ATTENDED.

```
SQL> select * from event_registration1 where PARTICIPATION_STATUS='ATTENDED';
```

REGISTRATION_ID	EVENT_ID	STUDENT_ID	REGISTRATION_DATE	PARTICIPATION_STATUS
ER101	EV101	S101	26-JUL-26	ATTENDED

Part D — Logical Operators, BETWEEN, IN, and LIKE

34. Display female students whose CGPA is greater than 8.

```
SQL> select * from student1 where gender='f' and cgpa >8;
```

STUDENT_ID	STUDENT_NAME	CGPA	CITY	STATE	CONTACT_NUMBER	DOB	EMAIL_ID	DEPT_ID
S103	Anjali Nair	9.6			9876512345	12-MAR-05	f_anjali@gmail.com	D01

35. Display students whose CGPA is between 7 and 9.

```
SQL> SELECT *
2 FROM STUDENT1
3 WHERE CGPA BETWEEN 7 AND 9;
```


STUDENT_ID	STUDENT_NAME	ADMISSION	CGPA	CITY	STATE	CONTACT_NUMBER	DOB	EMAIL_ID	DEPT_ID
S101	Akhil Raj	26-JUL-26	8.9			9876543210	15-AUG-05	ms101@gmail.com	D01

36. Display faculty members who are Assistant Professor and earn more than ₹45,000.

```
SQL> SELECT *
2 FROM FACULTY1
3 WHERE DESIGNATION = 'Assistant Professor'
4 AND SALARY > 45000;
```

FACULTY_ID	FACULTY_NAME	DESIGNATION	JOINING_D	SALARY	EMAIL_ID	DEPT_ID	EXPERIENCE_YEARS
F101	Meera Joseph	Assistant Professor	01-JUN-22	71280	meera@rajagiri.edu	D01	
F102	Rahul Menon	Assistant Professor	15-JUL-21	59400	rahul@rajagiri.edu	D01	

37. Display faculty members whose qualification is either PhD or MPhil.

Select * from faculty members where qualification is 'phd' or 'mphil';

38. Display faculty members whose salary is between ₹40,000 and ₹70,000.

```
SQL> select FACULTY_NAME from faculty1 where salary between 40000 and 70000;
```

FACULTY_NAME
Rahul Menon

39. Display courses offered in Semesters 1, 3, or 5 using the IN operator.

```
SQL> select * from course1 where semester IN(1,3,5);
```

COURSE_ID	COURSE_TITLE	CREDITS	SEMESTER	DEPT_ID	FACULTY_ID
C101	DBMS	5	3	D01	F102
C102	Operating Systems	4	3	D01	F102

40. Display courses not offered in Semesters 7 and 8.

SQL> select * from course1 where semester NOT IN(7,8);

COURSE_ID	COURSE_TITLE	CREDITS	SEMESTER	DEPT_ID	FACULTY_ID
C101	DBMS	5	3	D01	F102
C102	Operating Systems	4	3	D01	F102

41. Display results having grades A, B, or C.

SQL> select * from result1 where grade IN ('A','B','C');

RESULT_ID	ENROLLMENT_ID	INTERNAL_MARKS	EXTERNAL_MARKS	TOTAL_MARKS	G	RESULT_STA
R101	E101	28	45	73	B	PASS
R102	E102	35	42	77	B	PASS
R103	E103	40	48	88	A	PASS

42. Display events with registration fees between ₹100 and ₹1,000.

Select * from event1 where REGISTRATION_FEE between 100 and 1000;

EVENT_ID	EVENT_NAME	EVENT_DATE	VENUE	FACULTY_ID	MAX_PARTICIPANTS	REGISTRATION_FEE
EV101	Tech Fest	15-AUG-26	Main Auditorium	F101	100	500
EV103	Cultural Fest	10-OCT-26	Open Stage	F103	150	315

43. Display students whose names begin with the letter A.

SQL> select * from student1 where STUDENT_NAME like 'A%';

```
SQL> select * from student1 where STUDENT_NAME like 'A%';
```

STUDENT_ID	STUDENT_NAME	ADMISSION	CGPA	CITY	STATE	CONTACT_NUMBER	DOB	G EMAIL_ID	DEPT_ID
S101	Akhil Raj	26-JUL-26	8.9		Anjali Nair	9876543210	15-AUG-05 m	s101@gmail.com	D01
S102	Arun Kumar	26-JUL-26				9876545678	20-JAN-05 m	arun@gmail.com	D01
S103	Anjali Nair	26-JUL-26	9.6			9876512345	12-MAR-05 f	anjali@gmail.com	D01

44. Display students whose names contain the text Raj.

```
SQL> select * from student1 where STUDENT_NAME like '%Raj%';
```

```
SQL> select * from student1 where STUDENT_NAME like '%Raj%';
```

STUDENT_ID	STUDENT_NAME	ADMISSION	CGPA	CITY	STATE	CONTACT_NUMBER	DOB	G EMAIL_ID	DEPT_ID
S101	Akhil Raj	26-JUL-26	8.9			9876543210	15-AUG-05 m	s101@gmail.com	D01

45. Display faculty members whose email addresses belong to the rajagiri.edu domain.

```
SQL> select * from faculty1 where EMAIL_ID like '%rajagiri.edu';
```

```
SQL> select * from faculty1 where EMAIL_ID like '%rajagiri.edu';
```

FACULTY_ID	FACULTY_NAME	DESIGNATION	Is where C	JOINING_D	HR	SALARY	EMAIL_ID	DEPT_ID	EXPERIENCE_YEARS
F101	Meera Joseph	01 Assistant Professor	01-JUN-22	71280	meera@rajagiri.edu	D01			
F102	Rahul Menon	01 Assistant Professor	15-JUL-21	59400	rahul@rajagiri.edu	D01			
F103	Anitha Nair	02 Professor	10-JUN-20	77000	anitha@rajagiri.edu	D02			

46. Display courses whose course names contain the word Database.

```
SQL> SELECT *
2 FROM COURSE1
3 WHERE COURSE_TITLE LIKE '%Database%';
```

no rows selected

47. Display departments whose department names contain the word Science.

```
SQL> select * from department1 where DEPT_NAME like '%Science';
```

DEPT_ID	DEPT_NAME	OFFICE_LOCATION	CONTACT_NUMBER	HOD_FACULTY_ID
D01	Computer Science	Technology Block	9876543210	

Part E — NULL Values, Sorting, and Built-in Functions

48. Display students whose CGPA has not been entered.

```
SELECT *  
FROM STUDENT1  
WHERE CGPA IS NULL;
```

```
SQL> select * from student1 where cgpa is null; 48. Display courses for which no faculty member has been assigned.
```

STUDENT_ID	STUDENT_NAME ADMISSION	CGPA	CITY	STATE	CONTACT_NUMBER	DOB	G EMAIL_ID	DEPT_ID
S102	Arun Kumar 26-JUL-26				9876545678	20-JAN-05 m	arun@gmail.com	D01

```
SQL>
```

49. Display courses for which no faculty member has been assigned.

```
SQL> SELECT *  
2 FROM COURSE1  
3 WHERE FACULTY_ID IS NULL;  
  
no rows selected
```

50. Display all students in alphabetical order of student name.

```
SQL> select * from student1 order by STUDENT_NAME;
```

That's perfectly fine—it means all your current courses have a faculty member assigned.

STUDENT_ID	STUDENT_NAME ADMISSION	CGPA	CITY	STATE	CONTACT_NUMBER	DOB	G EMAIL_ID	DEPT_ID
S101	Akhil Raj 26-JUL-26	8.9			9876543210	15-AUG-05 m	s101@gmail.com	D01
S103	Anjali Nair 26-JUL-26	9.6			9876512345	12-MAR-05 f	anjali@gmail.com	D01
S102	Arun Kumar 26-JUL-26				9876545678	20-JAN-05 m	arun@gmail.com	D01

51. Display all students in descending order of CGPA.

```
SQL> select * from student1 order by cgpa desc ;
```

```
SQL> select * from student1 order by cgpa desc
```

STUDENT_ID	STUDENT_NAME	ADMISSION	CGPA	CITY	STATE	CONTACT_NUMBER	DOB	G E-MAIL_ID	DEPT_ID
S102	Arun Kumar	26-JUL-26				9876545678	20-JAN-05	m arun@gmail.com	D01
S103	Anjali Nair	26-JUL-26	9.6			9876512345	12-MAR-05	f anjali@gmail.com	D01
S101	Akhil Raj	26-JUL-26	8.9			9876543210	15-AUG-05	m s101@gmail.com	D01

52. Display faculty members in descending order of salary.

SQL> select * from faculty1 order by salary desc;

```
SQL> select * from faculty1 order by salary desc;
```

FACULTY_ID	FACULTY_NAME	DESIGNATION	JOINING_D	SALARY	E-MAIL_ID	DEPT_ID	EXPERIENCE_YEARS
F103	Anitha Nair	Professor	10-JUN-20	77000	anitha@rajagiri.edu	D02	
F101	Meera Joseph	Assistant Professor	01-JUN-22	71280	meera@rajagiri.edu	D01	
F102	Rahul Menon	Assistant Professor	15-JUL-21	59400	rahul@rajagiri.edu	D01	

53. Display courses first by semester and then by credits in descending order.

```
SQL> select semester , credits from course1 order by credits desc;
```

SEMESTER	CREDITS
3	5
3	4

54. Display all student names in uppercase.

SQL> select upper(student_name) from student1;

```
SQL> select upper(student_name) from student1;
```

UPPER(STUDENT_NAME)
AKHIL RAJ
ARUN KUMAR
ANJALI NAIR

55. Display faculty names with the first letter of each word capitalised using INITCAP.

```
SQL> SELECT INITCAP(faculty_name)
2 FROM faculty1;
```

```
SQL> SELECT INITCAP(faculty_name)
2 FROM faculty1;
```

INITCAP(FACULTY_NAME)
Meera Joseph
Rahul Menon
Anitha Nair

56. Display student names along with the number of characters in each name.

```
SELECT student_name, LENGTH(student_name)
FROM student1;
```

```
SQL> select length(student_name), student_name from student1;
```

LENGTH(STUDENT_NAME)	STUDENT_NAME
9	Akhil Raj
10	Arun Kumar
11	Anjali Nair

57. Display student names and CGPA values rounded to one decimal place.

```
SQL> SELECT student_name, ROUND(cgpa, 1)
2 FROM student1;
```

```
SQL> SELECT student_name, ROUND(cgpa, 1)
```

STUDENT_NAME	ROUND(CGPA,1)
Akhil Raj	8.9
Arun Kumar	9.2
Anjali Nair	9.6

58. Display faculty members and the year in which they joined.

SQL> select FACULTY_NAME, JOINING_DATE from faculty1;

```
SQL> select  FACULTY_NAME, JOINING_DATE  from faculty1;
```

FACULTY_NAME	JOINING_D
Meera Joseph	01-JUN-22
Rahul Menon	15-JUL-21
Anitha Nair	10-JUN-20

Part F — Aggregate Functions, GROUP BY, and HAVING

59. Write queries to find the total number of students.

SQL> select count(*) from student1 ;

```
SQL> select count(*) from student1 ;
```

COUNT(*)
3

60. Find the highest, lowest, and average CGPA of all students.

SQL> select max(cgpa), min(cgpa), avg(cgpa) from student1 ;

```
SQL> select max(cgpa), min(cgpa), avg(cgpa) from student1 ;
```

MAX(CGPA)	MIN(CGPA)	AVG(CGPA)
9.6	8.9	9.25